

L4 - Methods for Nondestructive Evaluation

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Signed: KS, AB,
, JN

Abstract

Five nondestructive evaluation (NDE) techniques—visual inspection, liquid penetrant, thermography, radiography, and ultrasonics—were examined in this lab to find surface and subsurface flaws without causing any damage to the components. A metallic wing airfoil's small surface fractures and scratches close to fastener holes were evident signs of damage at stress-concentration spots when examined under a borescope. Additional surface indications under UV lighting were revealed by liquid penetrant testing, confirming the dye/developer's sensitivity to shallow, tight fissures that were hardly noticeable to the human eye. Thermography using a FLIR camera demonstrated how thermal gradients can reveal nonuniformities in materials and bonding by clearly resolving temperature contrasts from an induced water chill and residual hand heat.

Radiography demonstrated the capacity to do volumetric inspection in situations where access is restricted by producing high-contrast interior pictures of a weld and a compact assembly. Echo time-of-flight versus position was mapped using ultrasonic testing on aluminum calibration blocks. Sensitivity to interior geometry was validated by tracking known thickness steps and internal ledges with decreasing echo spacing. When combined, the findings demonstrate complimentary strengths: depth-resolved flaw/thickness assessment (ultrasonics), rapid screening (visual), high surface sensitivity (penetrant), thermal anomaly detection (thermography), and internal visualization (radiography). For aerospace structures, where early detection at airfoil skins and joints is essential to preventing fatigue-driven damage progression, the integrated method offers a workable workflow.

Visual Inspection

Theory:

Visual inspection is one of the most basic nondestructive evaluation (NDE) techniques, which dates to the early days of industrial manufacturing when engineers could only detect flaws by sight. Due to its ease of use, affordability, and ability to yield quick results, it remains the most widely used NDE technique in aerospace maintenance, even with today's advanced technology.

It depends on the human eye's capacity to identify surface imperfections like dents, corrosion, cracks, and deformation before they become serious damage, often with the use of optical tools. Without damaging the material, this technique provides a quick and inexpensive evaluation of the structure. In aerospace applications, this can be considered as the first line of defense for quality control and maintenance to examine aircraft skins, joints, and interior parts. Borescopes, endoscopes, and magnifiers are examples of optical tools that increase visibility in difficult-to-reach places.

Visual inspection is frequently used in the aerospace industry to find early indications of corrosion, impact damage, or fatigue in wing airfoils, fuselage skins, fasteners, and joints. In this experiment, a borescope camera was used to examine a wing airfoil section.

This technique has several benefits, such as being inexpensive, providing quick feedback, and working with a variety of materials, including steel, composites, and aluminum. Some of the disadvantages are relying on human judgment, inability to identify surface flaws that might be hidden to the naked eye, and the need for proper lighting.

Equipment:

Below is the list of equipment used during the experiment:

- Ridgid Micro CA-150 Inspection Camera kit
- Funai LED display and remote control
- Video cable
- Extra AA batteries
- Imaging subjects (aircraft part)
- Instruction manual

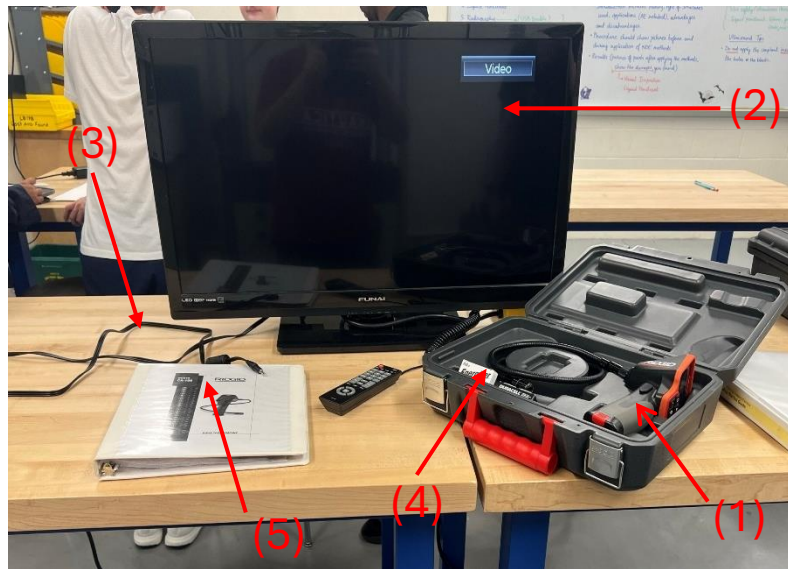


Figure 1: Apparatus Used: (1) - Ridgid Micro CA-150 Inspection Camera kit, (2) - Funai LED display and remote control, (3) - Video cable, (4) - Extra AA batteries, (5) - Instruction manual

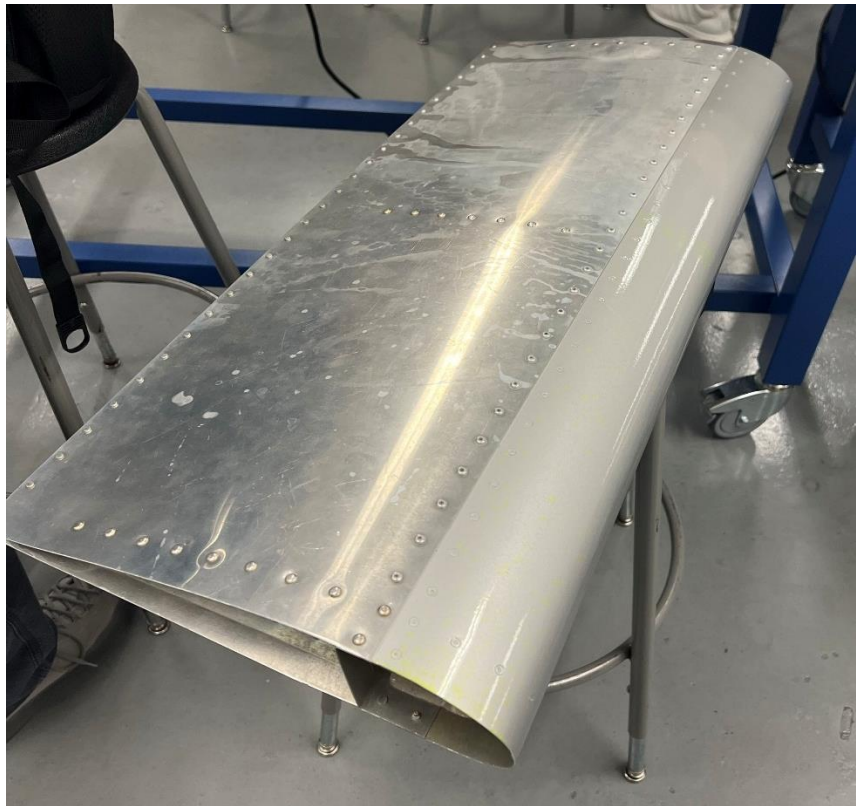


Figure 2: Imaging Subject – Aircraft Wing Airfoil

Method:

1. The inspection camera was powered on and connected to the LED display using the RCA cable.
2. The live feed was viewed by setting the display source to VIDEO mode.
3. External and internal surfaces of the aircraft's airfoil (testing object) were inspected for dents, corrosion, cracks, and manufacturing markings.
4. The borescope light was brightened for interior cavities and dimmed for external metallic surfaces to minimize glare.
5. Photographs were taken to document notable surface conditions and construction details.

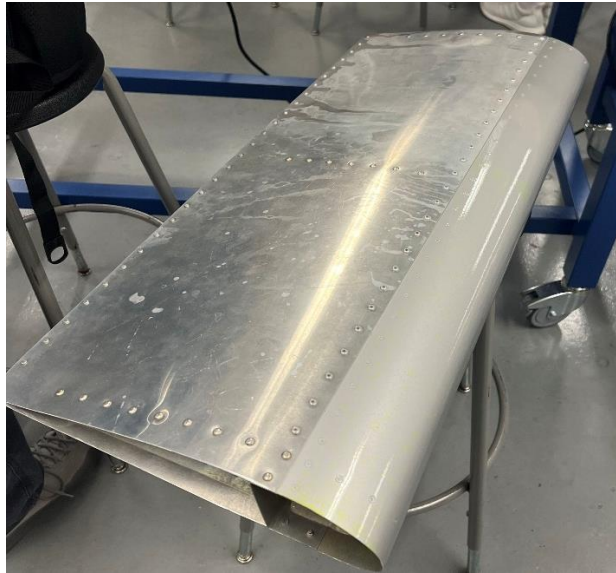


Figure 3: Before Inspection



Figure 4: During Inspection (Internal Surface of the Airfoil)

Results:

During inspection, the interior surface of the airfoil showed visible scratches and cracks, especially near the rivet holes. These damages are likely from handling, cyclic fatigue, or surface stress concentration. The borescope captured these features clearly and is presented in the following images. No corrosion or delamination was observed.



Figure 5: Interior Surface Deformations – Surface Crack



Figure 6: Interior Surface Deformations – Surface Crack



Figure 7: Interior Surface Deformations – Surface Scratches



Figure 8: Interior Surface Deformations – Surface Scratches

Analysis:

The damage found during the visual inspection is probably the result of fatigue damage. This is a common issue in wing structures when cyclic loading causes stress concentrations around fasteners. These surface defects are important because, if ignored, they might spread under repeated flight loads and ultimately result in material failure.

This lab demonstrated how visual inspection, particularly on vital aerodynamic structures like airfoils, can identify early-stage damage on accessible surfaces. However, follow-up techniques like liquid penetrant or ultrasonic inspection are necessary for a complete structural assessment because visual inspection is unable to detect underlying cracks.

Conclusion:

Visual examination of the airfoil successfully identified surface scratches and cracks, demonstrating the method's value in identifying early noticeable damage. While limited in depth detection, this technique remains vital for preliminary aircraft maintenance checks.

Thermography

Theory:

Thermography is one of the non-destructive evaluation techniques (NDE) that utilizes thermal imaging technology to detect changes in temperature by measuring the amount of infrared radiation emitted by an object. This technology was originally developed for the military for use in combat, its first steps occurring in the 1950s and 1960s by Texas Instruments, Hughes Aircraft, and Honeywell. Their crude designs would later be improved by Raytheon's R&D group in 1978, at the time, part of Honeywell. These early infrared detectors used barium strontium titanate (BST), which coated the thermal sensor. The year after this, Raytheon would go on to demonstrate this design to the military. Infrared imaging technology would eventually become commonplace, finding a place in law enforcement, security, search and rescue, and many other industries. Finally, in 1992, the American Society of Non-Destructive Testing would draft and release standards for courses involving thermal imaging. Common ways that thermography has been used in the aerospace industry are to observe materials subjected to heat; for example, observing thermal wear on jet engines after repeated use, wear that would otherwise be invisible to the naked eye. Another application is when inspecting welds; thermal imaging can expose defects in the weld, such as air bubbles. As previously mentioned, one of the benefits of thermography is that it can detect changes in a material that are invisible to the naked eye; however, that is not to say that thermography isn't without downsides. A big disadvantage of thermography is that only changes on the surface of a material are visible; any internal damage or deformations would remain undetected.

Equipment:

Equipment used for this lab is as follows:

1. FLIR T440 Thermal Imaging Camera
2. Instruction manual
3. Funai LED display and remote control
4. Imaging subjects (paper, water, hand imprint, etc.)

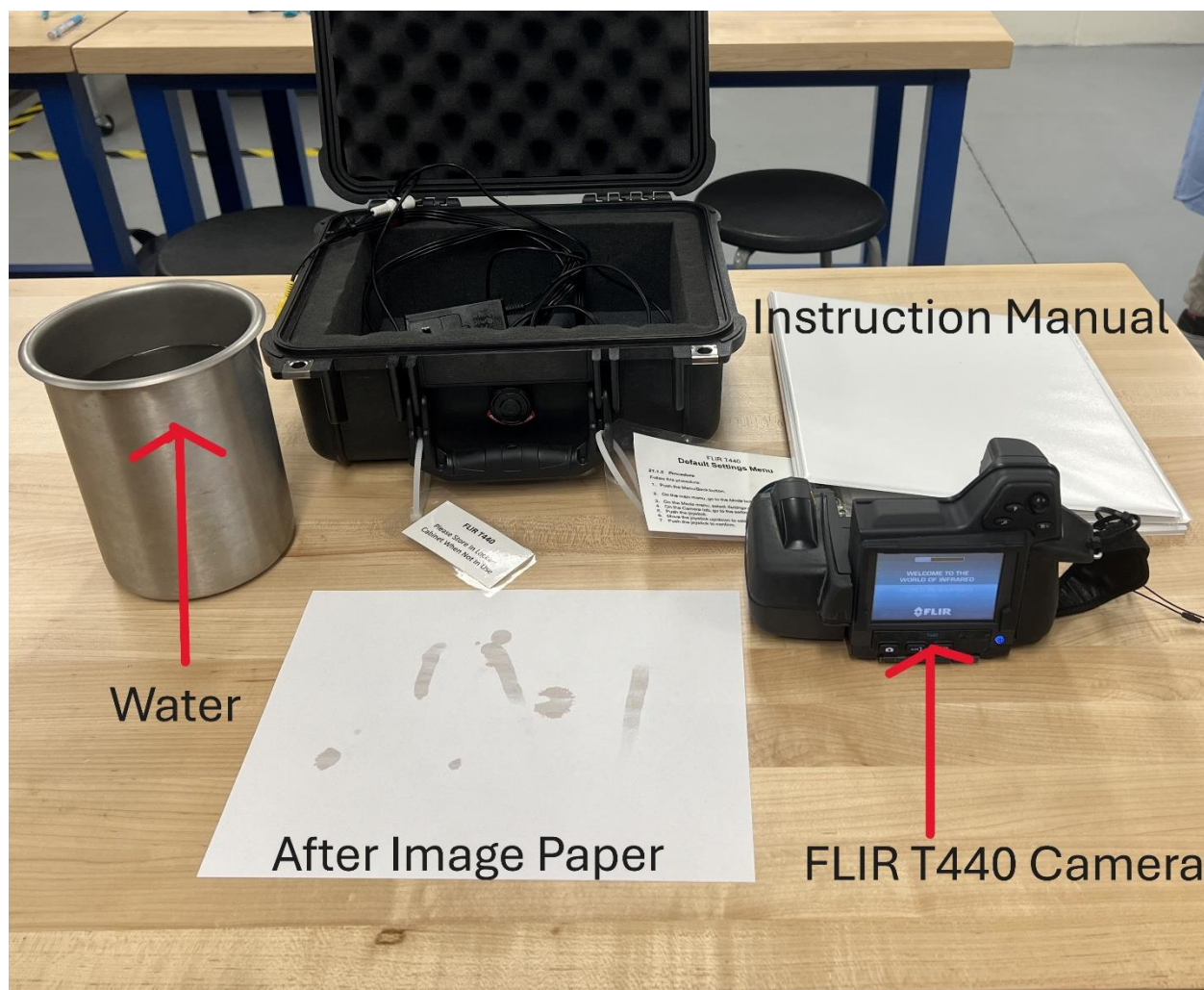


Figure 9: Equipment Used for Thermography

Method:

1. Start by reading the Instruction Manual to understand how to operate the FLIR T440 Camera.
2. Prep imaging subjects, paper being the subject for this experiment.

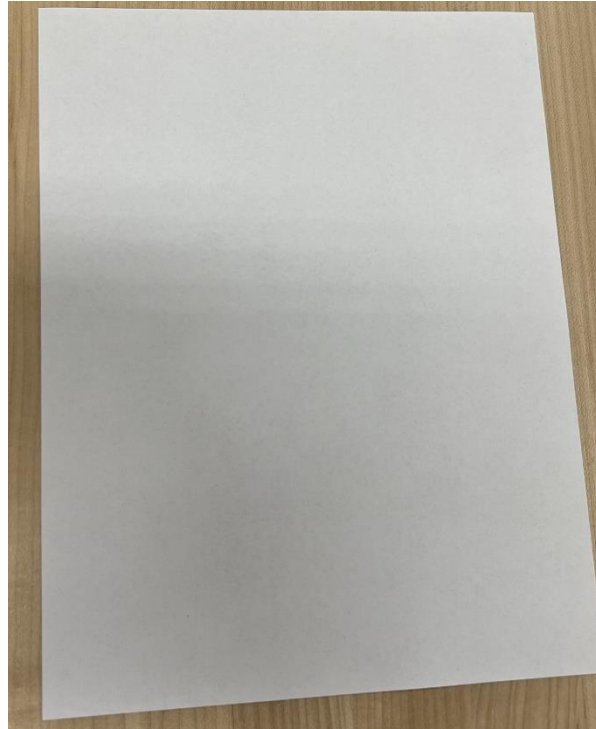


Figure 10: Image Subject Used

3. Take before images, with and without the FLIR camera, as shown in Figure 10.
4. Induce a temperature change in the image subject; water was chosen for this experiment, documenting this process such as shown in Figure 11.



Figure 11: Active Temperature Change in Image Subject

5. Take after images, with and without the FLIR camera, as shown below in figures 12 and 13.

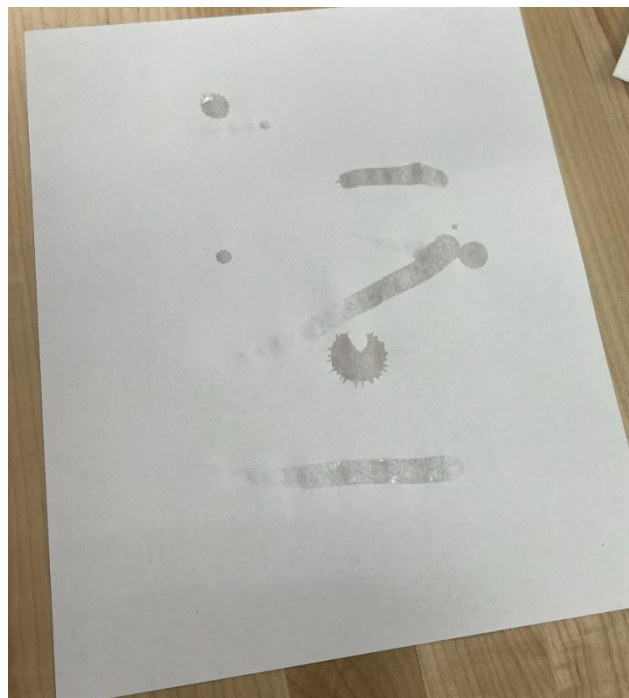


Figure 12: Image Subject After Inducing Temperature Change



Figure 13: Image Subject Under FLIR Camera After Temperature Change

6. Optionally, take images of residual heat; a hand imprint would be an example, as shown in Figure 13.

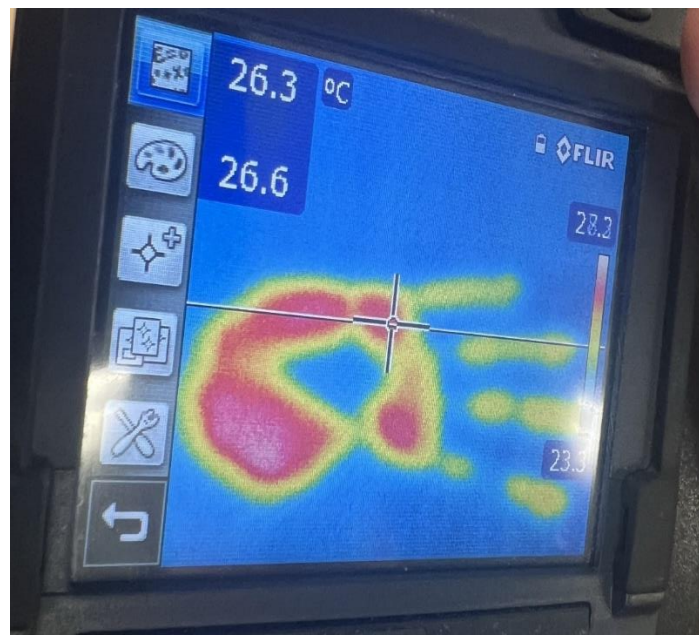


Figure 14: Residual Heat Left Behind From Hand Imprint

Results:

Figures 13 and 14 show how the FLIR camera picks up temperature changes, decreasing as shown in Figure 13 and increasing in Figure 14.

Discussion:

Thermography is a method that allows one to visually see infrared radiation given off by objects. Looking at Figure 14, the thermal camera could pick up the lingering heat left by a hand imprint. This would be important in aerospace as an option to observe gradual temperature changes over time, a quality that would point to how well a material disperses heat. Figure 13 shows that despite the camera's focus on thermal radiation, the absence of that thermal radiation can also be a sign of something amiss. Figure 13 could be what one would expect when observing a coolant spill, a drastic drop in temperature.

Conclusion:

Thermography is an important technique for NDE, as shown through this experiment. Thermography is a way to visually display changes in temperature when compared to its surroundings. By prepping an image subject and causing changes in temperature by adding splashes of water, comparing images before and after using the FLIR camera, we can see the temperature changes displayed visually, signified by darker spots on the camera. The significance of this is that thermography helps us see temperature changes when a material undergoes certain processes, such as thermal expansion, that could lead to thermal fatigue and an eventual critical failure of that part.

Radiography

Theory:

Radiography is a NDE method that allows the internal viewing of a components structure. It was discovered in 1895 by Wilhelm Conrad Roentgen using a cathode-ray tube. Two of his first experiments consist of a human hand and a box of weights. Most of the initial use of X-rays was in the medical field, but in 1913 high vacuum X-ray tubes were developed allowing for higher voltages than the normal cathode-ray tube. These higher voltages were necessary for observing through metal. A second method for radiography uses gamma rays instead of X-rays, first discovered in 1896. Gamma radiation utilizes materials that are considered radioactive like radium, cobalt, and iridium. This allows for much thicker objects to be observed but comes at the cost of being more dangerous.

Radiography is used on anything from bones to metal parts and has a wide variety of uses in many industries. In the aerospace industry it is primarily used to check the structural integrity of critical components like airframes and engines.

There are primary types of radiography used film, computed, digital, and computed tomography. Film is the oldest method and uses a film behind a object to capture the image. Computed utilizes a phosphor plate and scans the plate to convert it into a digital image. Digital uses a detector that converts it to a digital image and is at near real time. Computed tomography is similar to digital but creates a 3D image.

The advantages of radiography are primarily the viewing of internal structures, number of materials, the clarity of images, and its sensitivity. This method of NDE can be used on almost any material depending on the type of ray used and allows an image of the internal structure to be seen allowing it to be compared to the known structure. For its sensitivity it can detect a change of thickness of $\sim 1-2\%$. These do come at the disadvantages of high cost, low usefulness with high complexity parts, positioning, and safety concerns. The cost of radiography is much higher compared to other NDE methods but also raises significantly with the size of an object being tested due to a larger machine being used. While high complexity parts can be radiographed observing the resultant image becomes difficult as it needs to be compared to the desired result and can be hard to distinguish between discontinuities. The positioning of a radiographed part is important as both sides need to be accessed but also the way the rays pass through can cause different results as any discontinuity parallel to the ray may not be detected. Finally the safety of the operator is a large concern as radiography is using radiation, which is highly harmful to living creatures.

Equipment:

The following equipment is needed for Radiography:

- X-ray system
- Computer
- Test subject

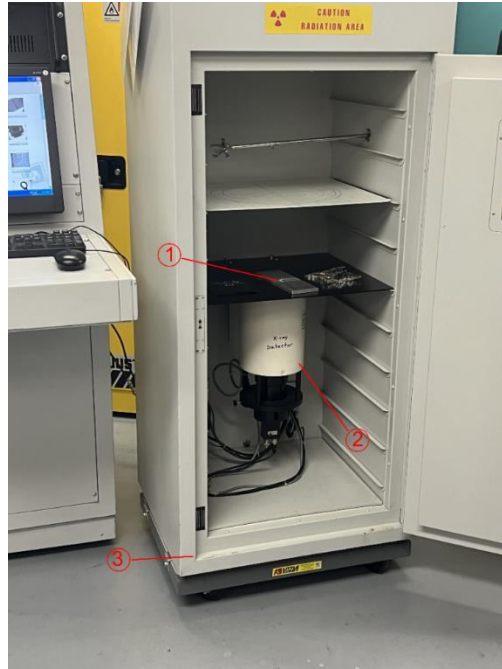


Figure 15: Radiography equipment: 1 – Test subject, 2 – X-ray detector, 3 – X-ray system



Figure 16: computer to partner with X-ray system

Method:

- 1) Run the imaging station software
- 2) Place subject in X-ray clamp
- 3) Close and secure X-ray station
- 4) Turn on X-ray station and adjust accelerating voltage until subject appears
- 5) Rotate subject until all desired angles are captured
- 6) Shut off X-ray station
- 7) Remove subject

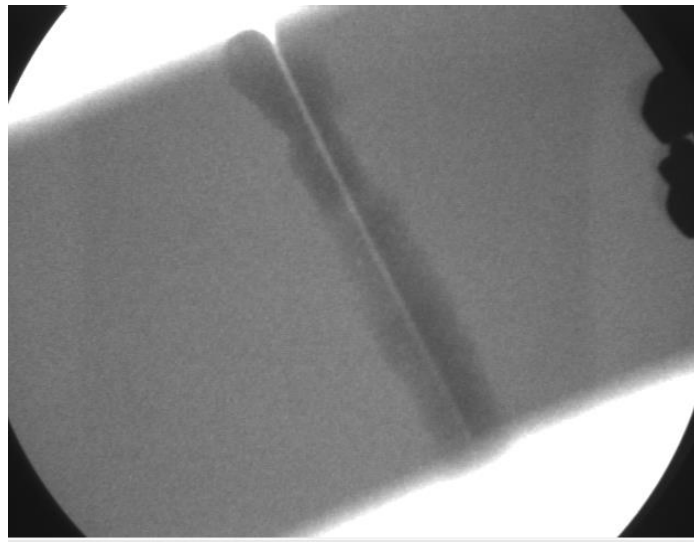
Results:

Figure 17: X-ray Inverse of a weld on an aluminum plate.

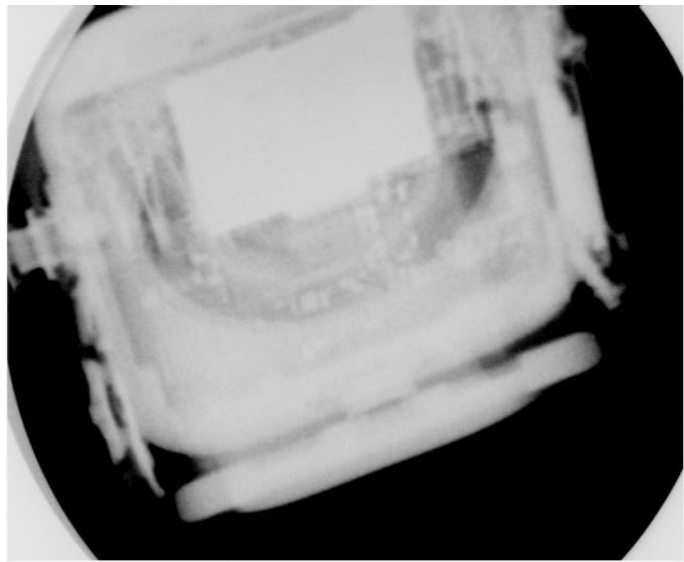


Figure 18: Xray of an apple watch

Discussion:

Utilizing radiography allows the inside structure to be observed. The two figures show both sides of how it is utilized. Figure 17 allows the structural analysis of a weld; useful to check if the weld meets industry standards and allows any defects to be observed. Figure 18 shows the internal components of the watch; useful to check that there are no defects on a system of parts without disassembling them. Figure 18 also shows the downside of radiography as while some components are observed the more complex structure causes issues with analyzing exact defects with parts overlapping.

Conclusion:

Radiography is a useful NDE technique but has many downsides when compared to other techniques. The cost and safety concerns compared to other methods are a major part of these downsides. However, the utility in seeing the internal structure of a component or weld offsets these concerns, as radiography is one of the only methods that visualizes these internal pieces without destroying the component.

Liquid Penetrant

Theory:

Liquid penetrant is one of the most popular NDE methods used across many industries. This popularity stems from its low cost and large testing ability. One of the first liquid penetrant examples is in the 1880s where oil and chalk was used to test railroad parts. In this a part was dipped in a discolored oil then dried and coated with a powdered chalk; the oil would then leak out of the cracks and discolor the chalk showing any defects. This process was furthered in the 1940s when dye was implemented into penetrating oils, showing any defects with the dyed locations.

In modern liquid penetrant there are three methods fluorescent particles, dyed particles, and a combination. Fluorescent particles are particles in a penetrating oil that glow when they are exposed to ultraviolet (UV) light. Dyed particles are dyes in the penetration oil, while a combination has both dyes and fluorescent particles in the oil. For all of these methods they work by having the penetrant sprayed on a surface letting it soak in for a period of time then wiping the surface. This should leave the penetrant in the cracks or indents and can be seen either with UV or normal light.

This method can be used with any non-porous material, leaving most metals rubber, plastics and glasses. In aerospace engineering this method is used on surface panels for aircraft along with most parts that are easily accessible and are known to ware. It is especially useful for aircraft skins as a large amount can be tested in a relatively short time.

The advantages of liquid penetrant are primarily its high sensitivity, its low cost, portability, and ease of use. Due to it being a liquid this method allows for damage not seen by the visual inspection to be found, but also highlights all surface damage so it is easily identifiable. It does come with disadvantages like only detecting surface damage, surface must be smooth and clean, and the chemicals used can be hazardous. Just like visual inspection liquid penetrant can only find damage if it is located on the surface of a part thus it also must be accessible by the inspector.

Equipment:

The equipment used in the experiment consist of:

- Cleaner
- Penetrant
- Developer
- Ultraviolet light
- Paper wiping cloth
- PPE
- Damaged part

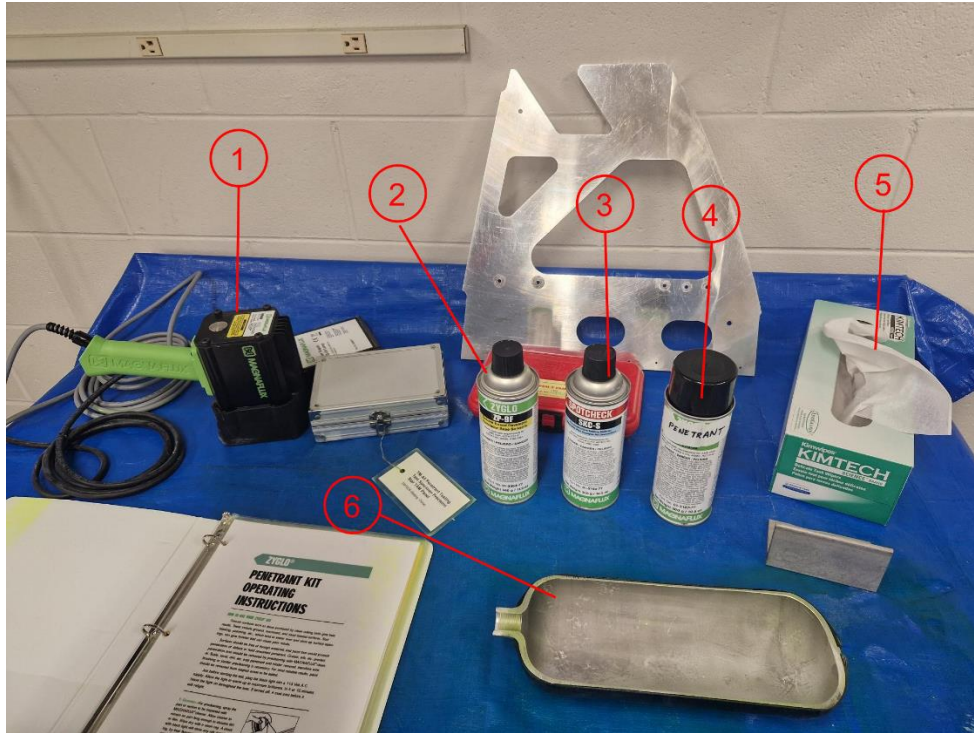


Figure 19: Liquid penetrate equipment: (1) – Ultraviolet light, (2) – Developer, (3) – Cleaner, (4) – Penetrant, (5) – Paper wipes, (6) – Test part.

Method:

1. Spray cleaner on test surface and let sit for ~30 sec. to allow oils to release from part, wipe clean
2. Spray a uniform layer of penetrant onto test surface; let sit for ~10 min.
3. Remove excess penetrant careful to not remove all
4. Spray developer in a uniform and light layer and let sit until dry.
5. Examine part using UV light to find damages in test surface.



Figure 20: Test Part before preparation.



Figure 21: Test surface with penetrant.



Figure 22: Test surface with penetrant and developer.

Results:

Figure 23: Test surface under UV light showing three damage areas.



Figure 24: Test surface under UV light showing five damage areas.

Discussion:

Using the liquid penetrant the damaged areas are more visible while utilising UV light. They contrast the green of the penetrant showing a brighter color. On the tank test part there are at least 8 sections that have damage. The damage that the penetrant shows is primarily scratches and cracks.

Conclusion:

Liquid penetrant is a useful tool to find surface level damage. Unlike visual inspection it highlights damage that may be hidden, allowing for small defects to be found. The main issues arise from damage that may not be surface level or the multi-step process compared to other testing methods.

Ultrasonics

Theory:

Ultrasonics testing is one of the most used nondestructive evaluation techniques, used for detecting internal flaws and issues, and measuring material thickness without causing damage to the structure. This method works through the propagation of high-frequency mechanical waves through solid material. The behavior of these waves is then analyzed as they change in the material's internal structure.

The first Ultrasonics testing device was developed by Dr. Floyd Firestone at the University of Michigan in 1940. The device was patented as the "Supersonic Reflectoscope". Dr. Firestone's design used the same principle as a sonar, it sent a pulse of high-frequency sound into a solid and the user would then observe the reflected echoes from flaws or boundaries. Since the creation of the device, it has become widely used in aerospace, automotive, civil engineering fields, as well as medical fields where ultrasound imaging is used to help a physician to evaluate, diagnose and treat medical conditions. In engineering fields, Ultrasonics testing is used to ensure the integrity of components like turbine blades, aircraft wings and welds.

The way it works is that a piezoelectric transducer converts electrical energy into mechanical vibrations that travel through a tested structure as longitudinal waves. Whenever the waves come against a surface like a crack, a void or just a back wall of the material, part of the energy is reflected back to the transducer. The Ultrasonics Inspection Device then converts the returning mechanical energy back into an electrical signal, appearing on the display as a peak or echo.

The purpose of this experiment was to get familiar with the Ultrasonics Testing and observe how ultrasonic waves behave when travelling through aluminum blocks. The effects of internal features of the blocks such as holes and ledges on the reflected signals on the displays were analyzed. This showed how Ultrasonics Testing can reveal hidden anomalies that would otherwise be invisible to the naked eye.

The advantages of Ultrasonics Testing are: it can detect very small internal flaws, can be performed on one side of the part and works well for metals, composites and ceramics which are commonly used in the aerospace industry. The disadvantages are: ultrasonic testing needs a smooth and clean surface that was well coated in an ultrasonic coupling and a skilled interpretation of the results.

Despite these limitations, the precision and reusability Ultrasonic Testing provides make it one of the most valuable tools for ensuring the structural integrity in the aerospace industry.

Equipment:

Below is the list of equipment that was used in this experiment:

- Portable ultrasonic inspection device
- 5 MHz piezoelectric transducer and cable
- Ultrasonic couplant
- Aluminum calibration blocks (SCB)
- Other test blocks
- Instruction Manual
- Task Wipers
- Cotton Swabs

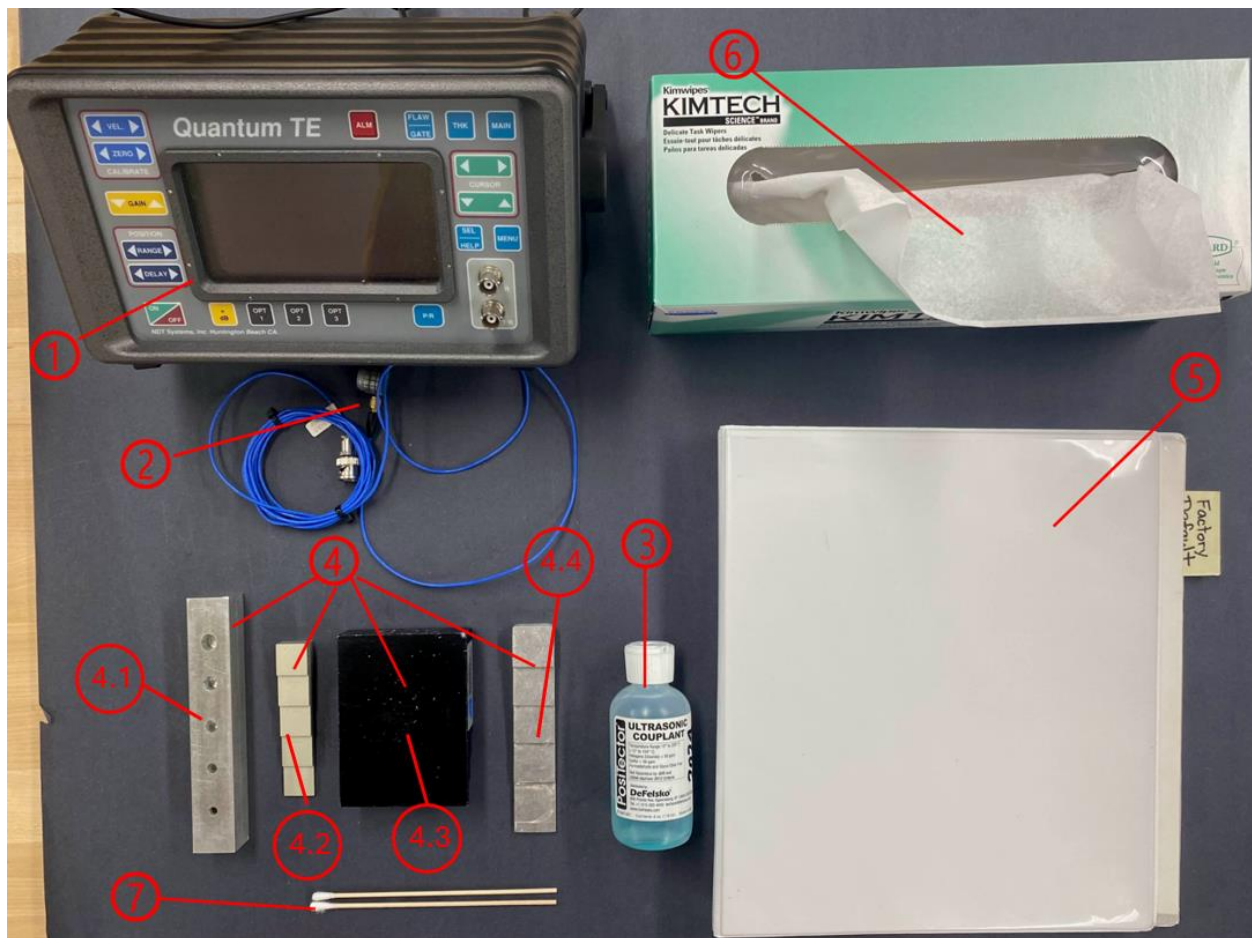


Figure 25: Ultrasonics equipment: (1) – Portable Ultrasonic Inspection Device, (2) – 5MHz piezoelectric transducer and cable, (3) – Ultrasonic Couplant, (4.1-4.4) – Aluminum Calibration Blocks, (5) – Instruction Manual, (6) – Paper wipes, (7) – Cotton Swabs

Method:

1. The transducer was connected to the inspection device
2. The inspection device was turned ON, the Gain and Range were adjusted to approximately 50
3. A small amount of couplant was applied to the calibration block and was spread uniformly using a cotton bud
4. The transducer was placed on the flat surface of the calibration block and moved around slowly
5. While moving the block, pictures of the screen output of the portable ultrasonic inspection device were taken depending on the inspection block features such as ledges and holes
6. If problems arose, the ultrasonic inspection device was reset to default settings
7. The portable ultrasonic inspection device was turned OFF and the working station and equipment were cleaned up for the next groups to use

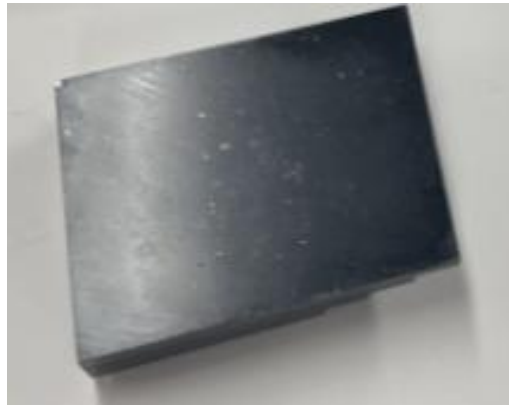


Figure 26: Aluminum Calibration Block before the experiment

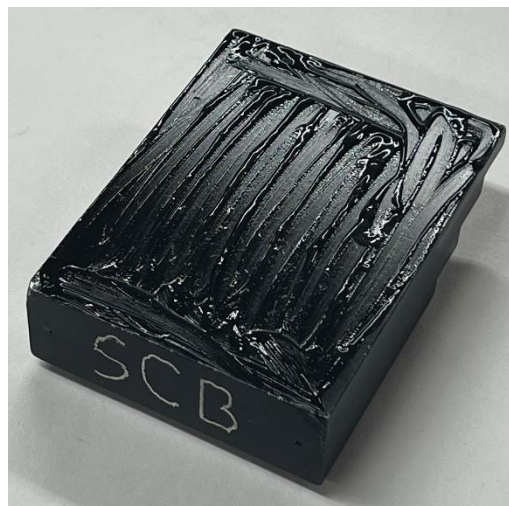


Figure 27: Aluminum Calibration Block after the application of Ultrasonic Couplant



Figure 28: Aluminum Calibration Block during the experiment

Results:

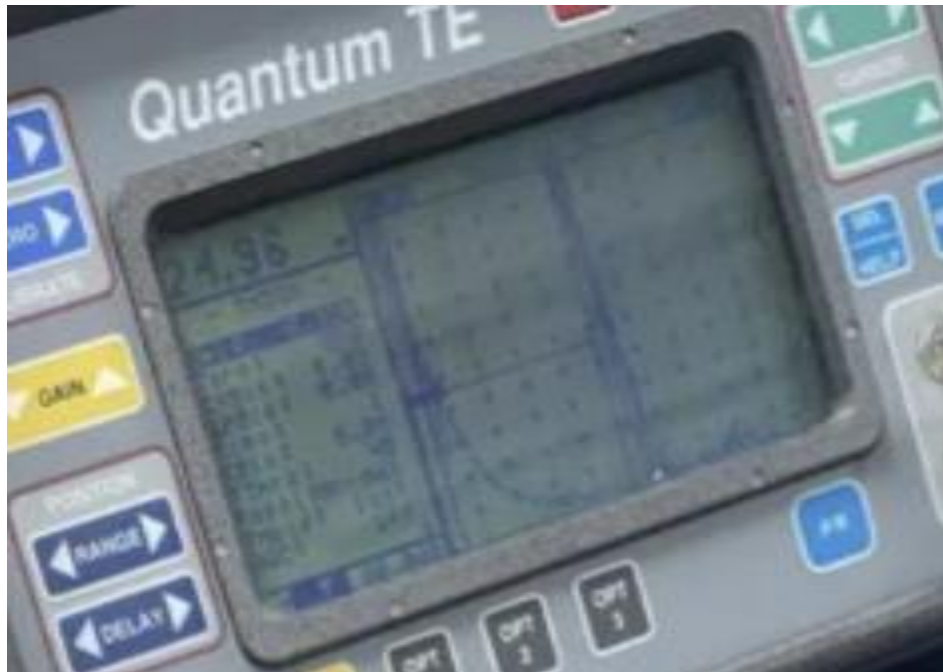


Figure 29: Ultrasonic inspection device screen for showing the echo pattern for Aluminum Calibration Block 4.3 at the tallest section of the sample



Figure 30: Ultrasonic inspection device screen for showing the echo pattern for Aluminum Calibration Block 4.3 at the second tallest section of the sample



Figure 31: Ultrasonic inspection device screen for showing the echo pattern for Aluminum Calibration Block 4.3 at the second smallest section of the sample



Figure 32: Ultrasonic inspection device screen for showing the echo pattern for Aluminum Calibration Block 4.3 at the smallest section of the sample

Discussion:

Figures 29-32 illustrate a progression of echo patterns as the transducer was moved from the regions of highest to lowest thickness. As the material thickness decreases, the time spacing between these peaks also decreases. The aptitude of echoes slightly changes, because of attenuation and partial scattering at ledges of the block.

The height of each echo corresponds to the intensity of the reflected sound. Consistent peaks mean smooth and continuous surfaces and smaller and distorted peaks mean anomalies like edges that scatter the wave energy.

This principle is used in aerospace applications to detect cracks, corrosions, and delaminations in metallic and composite materials. A similar procedure is used when inspecting aircraft skins or turbine blades.

Conclusion:

Ultrasonic inspection effectively showed the internal features of the aluminum test block. The procedure showed the method is accurate and repeatable for detecting hidden flaws in aerospace materials. The main drawback of this Nondestructive Evaluation (NDE) is the dependence on the surface condition of the material but overall it remains as one of the most reliable volumetric NDE methods.

Acknowledgements

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A. Bharadwaj: Visual Inspection

J. Nguyen Jr.: Thermography

R. Orr: Liquid Penetrant, Radiography

K. Satvaldy: Ultrasonics

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